



Introduction

Fraction means 'part or fragment of a whole object or collection'.

It is based on the concept of 'fair sharing' or 'equal division'.

Thus fraction denotes 'part(s) out of equal parts of a whole'.

One-half

When an object is divided into two equal parts, then each part is called **one-half** of the whole.

Divide a circle into 2 equal parts.

Shade 1 part.

The shaded part is 1 part out of 2 equal parts.

The shaded part is called one-half.

We express **one-half** by $\frac{1}{2}$ and read it as **one by two** or **one over two**.

The unshaded part is also 1 part out of 2 equal parts. So, the unshaded part is also one-half.



Remember:

To express as a fraction, equal division is a must.

Look at the following figures:



Figure A

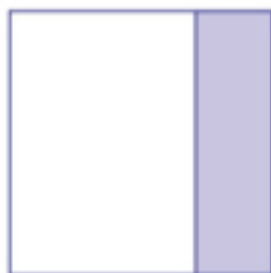


Figure B



Figure C



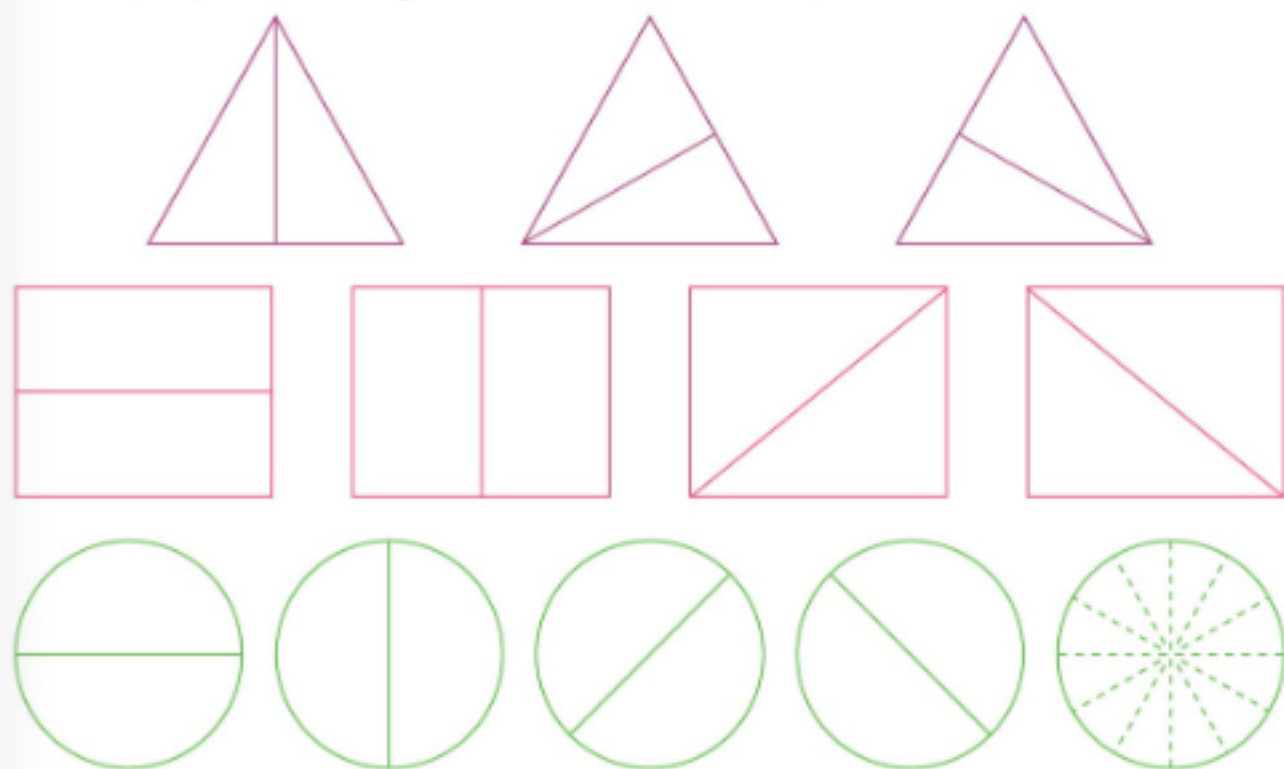
Figure D

Each of the figures A, B, C, and D is divided into 2 parts.

But the shaded part in Figure A only shows one-half of the whole, as the two parts in Figure A are equal.

In Figures B, C, and D, the two parts are not equal. The shaded part in Figure B and Figure C is less than one-half and the shaded part in Figure D is more than one-half of the whole.

There are many ways in which a figure can be divided into two equal halves:



One-third, two-thirds

When an object is divided into three equal parts, then each part is called **one-third** of the whole.

Divide a circle into 3 equal parts. Shade 1 part.

The shaded part is 1 part out of 3 equal parts.

The shaded part is called one-third.

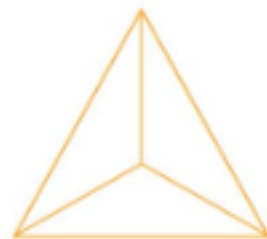
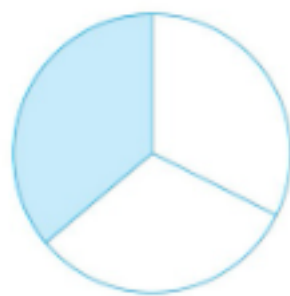
We express **one-third** by $\frac{1}{3}$ and read it as **one by three** or **one over three**.

The unshaded part is 2 parts out of 3 equal parts. We say that the unshaded part is two-thirds of the whole.

We express **two-thirds** by $\frac{2}{3}$ and read it as **two by three** or **two over three**.

In the figure shown alongside, a triangle with all sides equal, is divided into three equal parts.

Can you think of more ways to do so? Say 'yes' or 'no'.



Can you divide a triangle with unequal sides, into three equal parts?

State 'yes' or 'no'.



One-fourth, two-fourths, three-fourths

When an object is divided into four equal parts, then each part is called **one-fourth (or quarter)** of the whole.

Divide a circle into 4 equal parts. Shade 1 part.

The shaded part is 1 part out of 4 equal parts.

The shaded part is called one-fourth or quarter of the whole.

We express **one-fourth** by $\frac{1}{4}$ and read it as **one by four** or **one over four**.

The unshaded part is 3 parts out of 4 equal parts.

The unshaded part is, thus, three-fourths or three-quarters of the whole.

We express **three-fourths** by $\frac{3}{4}$ and read it as **three by four** or **three over four**.

In the adjoining figure, 2 parts out of 4 equal parts are shaded.

So, the shaded part is **two-fourths** or two-quarters of the whole, written as $\frac{2}{4}$.

Look at the following figures:

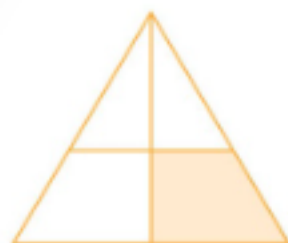


Figure A

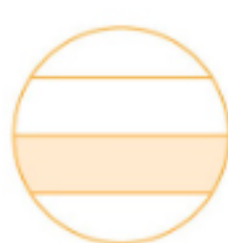


Figure B



Figure C



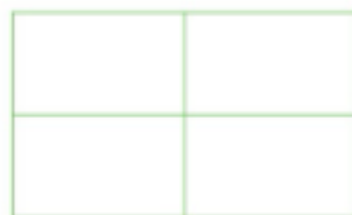
Figure D

Again, we can see that only Figure C is divided into 4 equal parts. In Figures A, B and D, the parts are unequal. So, the shaded part represents one-fourth in Figure C only.

Look at the following rectangle.



How many different ways can you think of to divide the above rectangle into 4 equal parts? Here are some of them:



What do you have to say about  ?

No, in this case, the four parts are not equal.

One-fifth, two-fifths etc.

Take a rectangular paper strip and divide it into 5 equal parts as shown. Shade 1 part.

The shaded part is one-fifth of the whole strip, written as $\frac{1}{5}$.



The unshaded part is 4 parts out of 5 equal parts. So, the unshaded part is four-fifths of the whole strip, written as $\frac{4}{5}$.

Now, look at the paper strip shown below.



Here, the shaded part is 2 parts out of 5 equal parts.

So, the shaded part is two-fifths of the whole strip, written as $\frac{2}{5}$.

The unshaded part is 3 parts out of 5 equal parts.

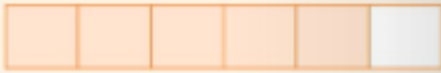
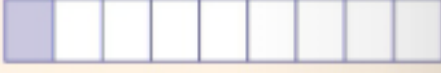
It is thus, three-fifths of the whole strip, written as $\frac{3}{5}$.



Fractional Numbers and Fractions

The numbers such as one-half, one-third, two-thirds, one-fourth, two-fourths, three-fourths and two-fifths etc. are known as **fractional numbers** and their symbols $\frac{1}{2}$, $\frac{1}{3}$, $\frac{2}{3}$, $\frac{1}{4}$, $\frac{2}{4}$, $\frac{3}{4}$, $\frac{2}{5}$ etc. are known as **fractions**.

Some more examples are given below:

	Shaded parts	Fraction	Fractional number	Figure
(i)	3 parts out of 5 equal parts	$\frac{3}{5}$	Three-fifths	
(ii)	5 parts out of 6 equal parts	$\frac{5}{6}$	Five-sixths	
(iii)	4 parts out of 7 equal parts	$\frac{4}{7}$	Four-sevenths	
(iv)	3 parts out of 8 equal parts	$\frac{3}{8}$	Three-eighths	
(v)	1 part out of 9 equal parts	$\frac{1}{9}$	One-ninth	

How many halves are there in a whole?

Divide a circle into 2 equal parts.

Then, each part is one-half.

$\therefore$ Two halves make a whole.

Now, 2 parts out of 2 equal parts make a whole.

$$\therefore \frac{2}{2} = \text{whole} = 1.$$



Two-halves or whole

How many one-thirds are there in a whole?

Divide a circle into 3 equal parts.

Then, each part is one-third.

Now, three-thirds make a whole.

Thus, 3 parts out of 3 equal parts make a whole.

$$\therefore \frac{3}{3} = \text{whole} = 1.$$



Three-thirds or whole

How many one-fourths are there in a whole?

Divide a circle into 4 equal parts.

Then, each part is one-fourth.

Here, four-fourths make a whole.

Thus, four parts out of 4 equal parts make a whole.

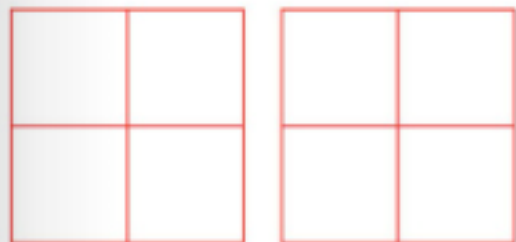
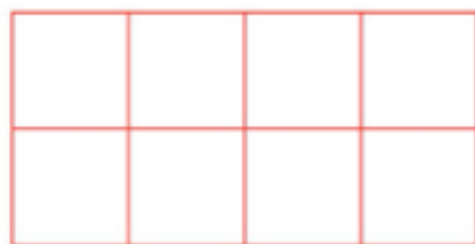
$$\therefore \frac{4}{4} = \text{whole} = 1.$$



Four-fourths or whole

Half and Double

Rohan has 8 blocks as shown.



Sohan comes and takes 4 blocks.

Clearly, Sohan takes one-half of the blocks.

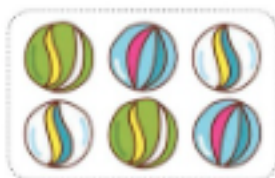
We say that 4 is half of 8;

and 8 is double of 4.

Anu has 3 marbles.



Tanu has 6 marbles.



We can say that Anu has half the number of marbles that Tanu has.

And, Tanu has double the number of marbles that Anu has.

Similarly, 5 is half of 10;

And, 10 is double of 5.

and so on.



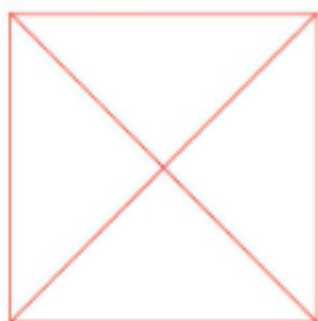
Exercise 39

1. Tick (✓) the figures which are divided into equal parts.

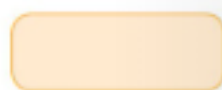
(a)



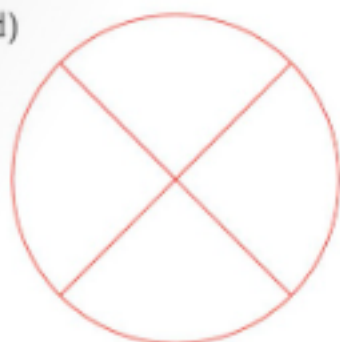
(b)



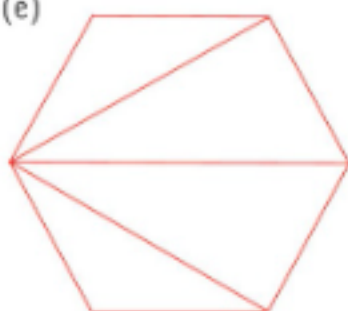
(c)



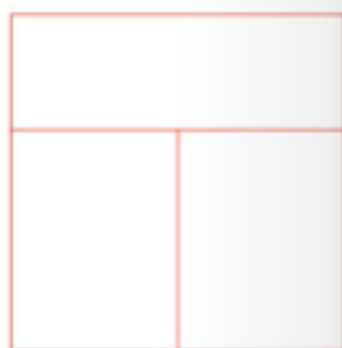
(d)



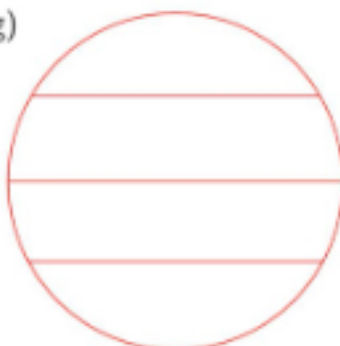
(e)



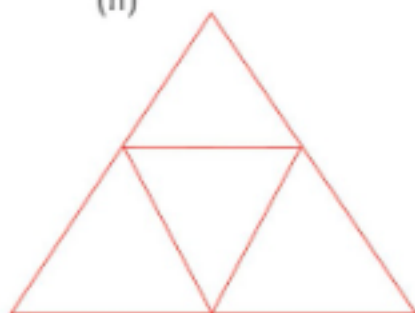
(f)



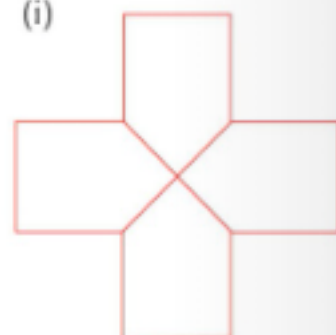
(g)



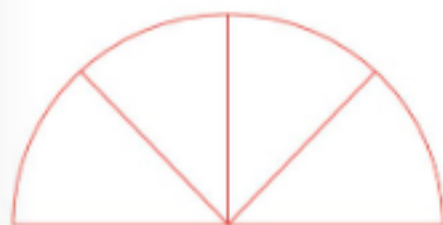
(h)



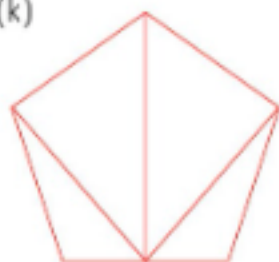
(i)



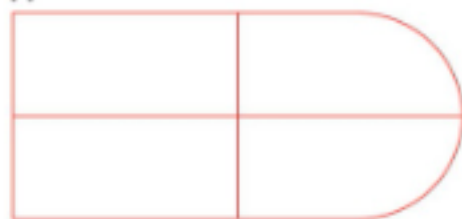
(j)



(k)



(l)

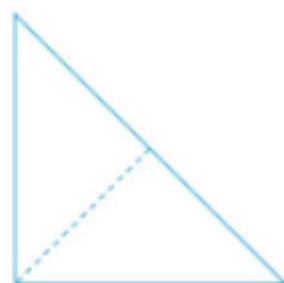


(m)

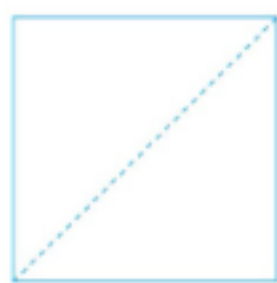


2. Shade with a pencil one-half or $\frac{1}{2}$ of each of the following figures.

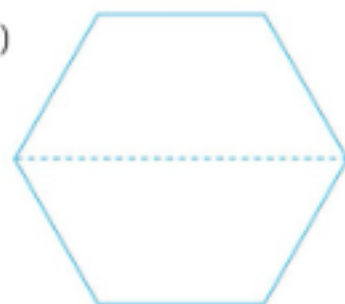
(a)



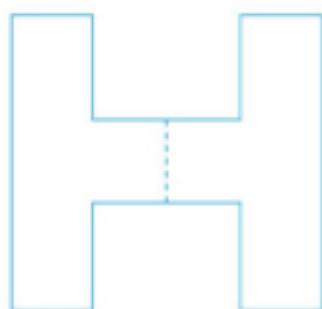
(b)



(c)



(d)



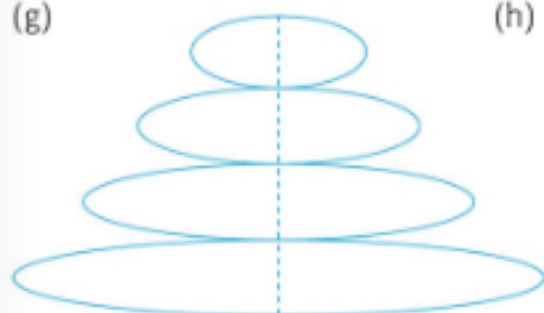
(e)



(f)



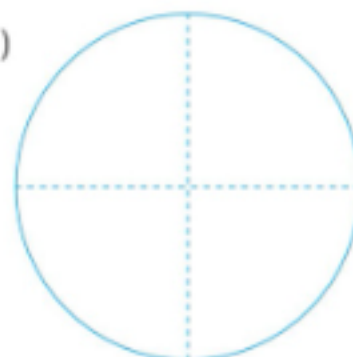
(g)



(h)



(i)



3. Shade with a pencil one-third or $\frac{1}{3}$ of each of the following figures.

(a)



(b)



(c)



(d)



(e)

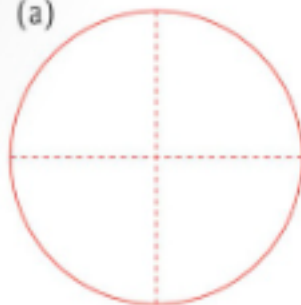


(f)



4. Shade with a pencil one-fourth or $\frac{1}{4}$ of each of the following figures.

(a)



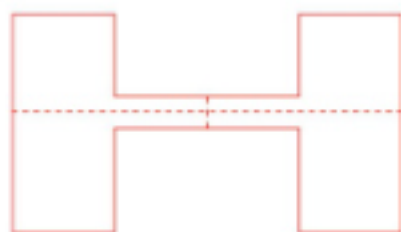
(b)



(c)



(d)



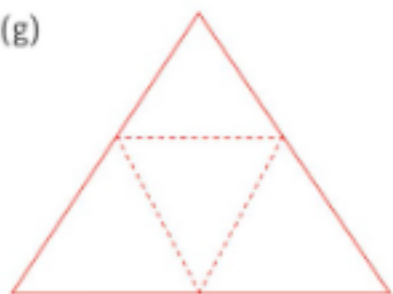
(e)



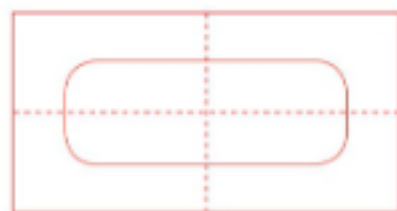
(f)



(g)



(h)

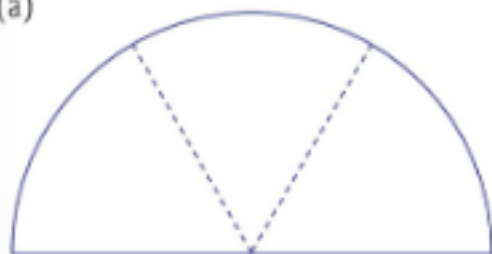


(i)



5. Shade with a pencil two-thirds or $\frac{2}{3}$ of each of the following figures.

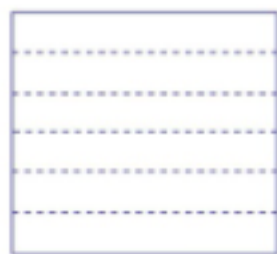
(a)



(b)

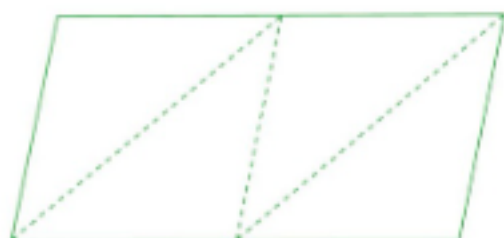


(c)

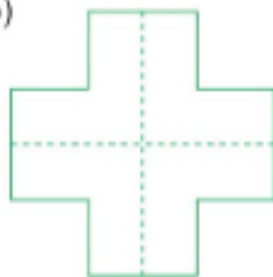


6. Shade with a pencil three-fourths or $\frac{3}{4}$ of each of the following figures.

(a)



(b)

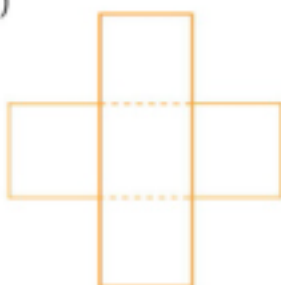


(c)

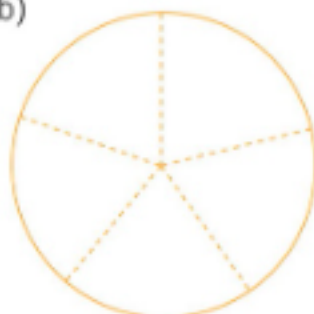


7. Shade with a pencil $\frac{3}{5}$ part of each of the following figures.

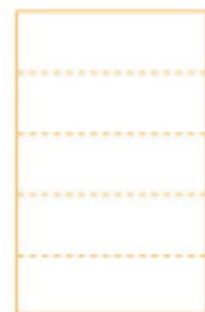
(a)



(b)

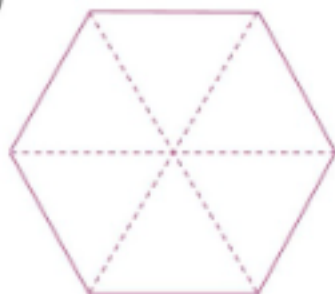


(c)

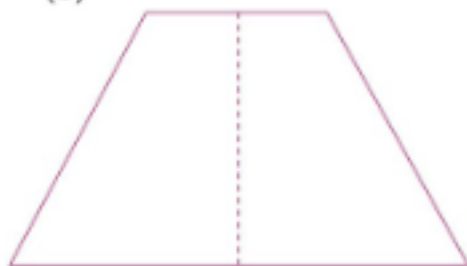


8. Shade a part of each whole to represent the given fraction.

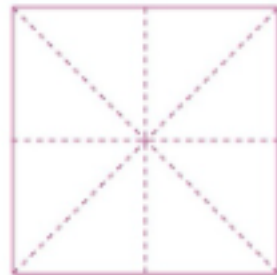
(a)



(b)



(c)



$$\frac{1}{6}$$

$$\frac{1}{2}$$

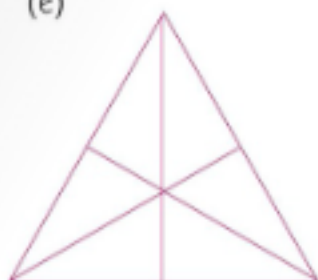
$$\frac{3}{8}$$

(d)



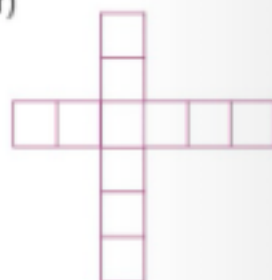
$$\frac{4}{9}$$

(e)



$$\frac{3}{6}$$

(f)



$$\frac{6}{11}$$

(g)



$$\frac{4}{9}$$

(h)



$$\frac{3}{8}$$

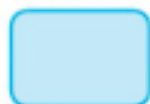
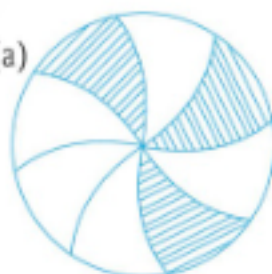
(i)



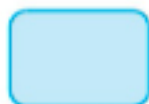
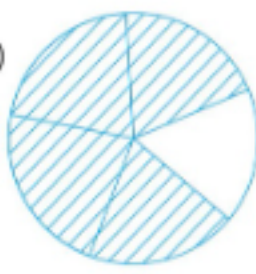
$$\frac{5}{11}$$

9. For each of the following figures write the fraction shown by the shaded parts.

(a)



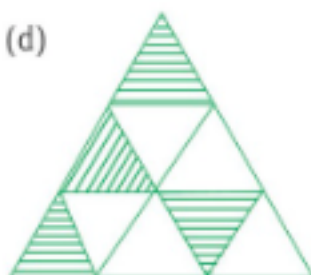
(b)



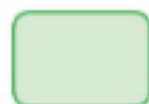
(c)



(d)



(e)

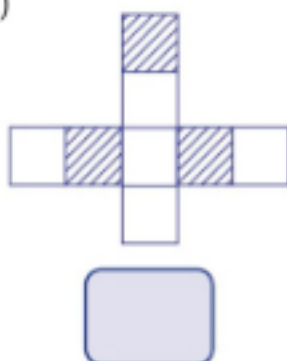


(f)

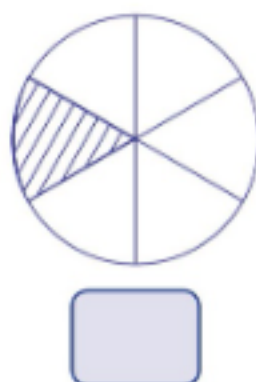


10. For each of the following write the fraction shown by the unshaded parts.

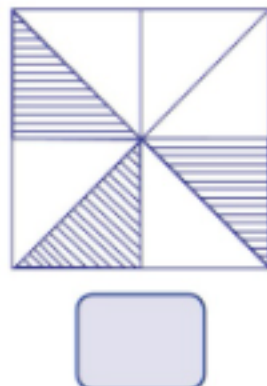
(a)



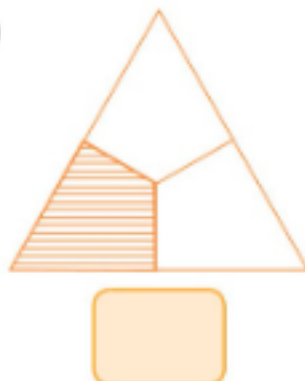
(b)



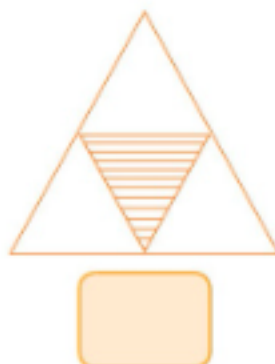
(c)



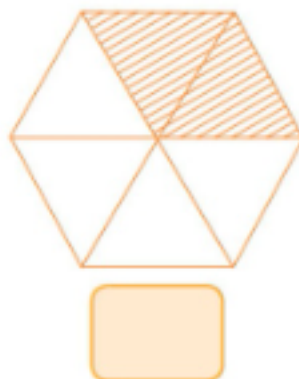
(d)



(e)



(f)



(g)



11. Write the fraction for each of the following fractional numbers. One has been done for you.

Fractional Number		Fraction	Fractional Number		Fraction
(a)	three-sevenths	$\frac{3}{7}$	(b)	one-fifth	
(c)	three-sixths		(d)	five-sixths	
(e)	one-eighth		(f)	seven-tenths	
(g)	six-ninths		(h)	nine-elevenths	
(i)	eight-twelfths		(j)	twelve-seventeenths	

12. Write the fractional number for each of the following fractions. One has been done for you.

Fraction		Fractional Number	Fraction		Fractional Number
(a)	$\frac{4}{8}$	four-eighths	(b)	$\frac{5}{7}$	
(c)	$\frac{1}{5}$		(d)	$\frac{4}{6}$	
(e)	$\frac{7}{9}$		(f)	$\frac{6}{11}$	
(g)	$\frac{8}{10}$		(h)	$\frac{9}{12}$	
(i)	$\frac{2}{3}$		(j)	$\frac{7}{16}$	

Numerator and Denominator of a Fraction

A fraction is written with two numerals arranged one over the other and separated by a line. The numeral above the line is called the **numerator** and the numeral below the line is called the **denominator** of the fraction.

In $\frac{2}{3}$, we have numerator = 2, denominator = 3.

In $\frac{4}{7}$, we have numerator = 4, denominator = 7.

In $\frac{1}{8}$, we have numerator = 1, denominator = 8.



Exercise 40



1. Write the numerator and denominator of each of the following fractions.

	Fraction	Numerator	Denominator
(a)	$\frac{1}{7}$		
(c)	$\frac{2}{9}$		
(e)	$\frac{11}{24}$		
(g)	$\frac{17}{35}$		
(i)	$\frac{7}{23}$		

	Fraction	Numerator	Denominator
(b)	$\frac{3}{11}$		
(d)	$\frac{7}{10}$		
(f)	$\frac{8}{13}$		
(h)	$\frac{10}{30}$		
(j)	$\frac{22}{41}$		

2. Write the fractions in which:

- | | |
|-------------------------------------|--------------------------------------|
| (a) Numerator = 7, Denominator = 12 | (b) Numerator = 5, Denominator = 8 |
| (c) Numerator = 4, Denominator = 5 | (d) Denominator = 7, Numerator = 2 |
| (e) Numerator = 3, Denominator = 6 | (f) Denominator = 10, Numerator = 9 |
| (g) Denominator = 15, Numerator = 7 | (h) Denominator = 25, Numerator = 14 |

3. Fill in the blanks.

- | | |
|---|---|
| (a) In $\frac{10}{17}$, the is 17. | (b) In $\frac{8}{15}$, the is 8. |
| (c) In $\frac{14}{19}$, the numerator is | (d) In $\frac{17}{23}$, the denominator is |
| (e) In $\frac{16}{21}$, the is 16 and the is 21. | |

Fractional Part of a Collection or Group

As we have studied earlier,

$\frac{1}{2}$ means '1 part out of 2 equal parts';

$\frac{1}{3}$ means '1 part out of 3 equal parts';

$\frac{1}{4}$ means '1 part out of 4 equal parts';

$\frac{2}{5}$ means '2 parts out of 5 equal parts';

and so on.

The above fact can be used to shade or find the fractional part of a collection or group of objects.

Let us find and shade $\frac{1}{2}$ of a collection of 6 balls.

We divide the collection into two equal parts by a line as shown.

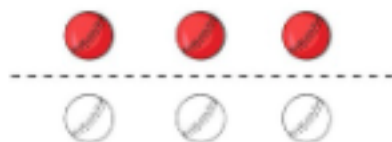
We now shade the balls in one part.

The shaded part represents $\frac{1}{2}$ of the collection. Clearly, $\frac{1}{2}$ of 6 balls = 3 balls.

Thus, $\frac{1}{2}$ of 6 = 3.

Also, $6 \div 2 = 3$.

So, to find $\frac{1}{2}$ of a collection, we divide by 2.



Let us now find and shade $\frac{1}{3}$ of a collection of 12 leaves.

We divide the collection into 3 equal parts by drawing lines as shown.

We now shade the leaves in one part.

The shaded part represents $\frac{1}{3}$ of the collection.

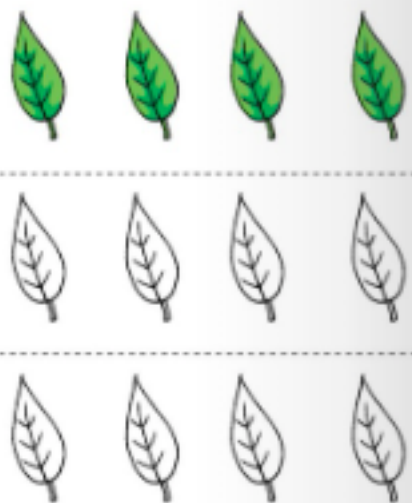
Clearly, $\frac{1}{3}$ of 12 leaves = 4 leaves.

Thus, $\frac{1}{3}$ of 12 = 4.

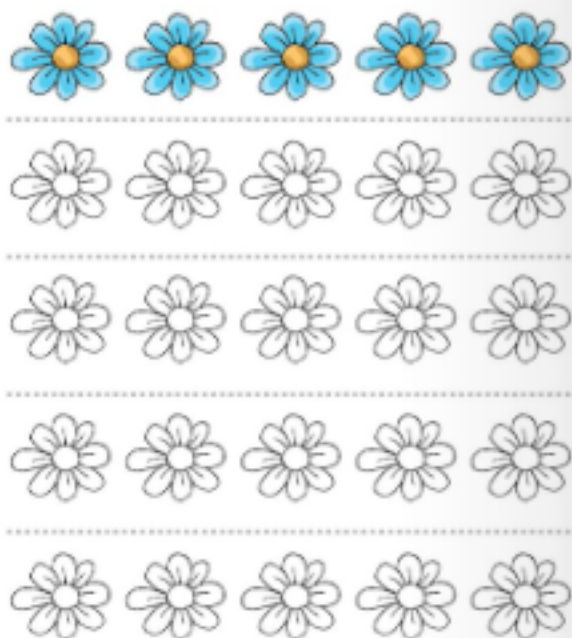
Also, $12 \div 3 = 4$.

So, to find $\frac{1}{3}$ of a collection, we divide by 3.

Similarly, we may find $\frac{1}{4}$ or $\frac{1}{5}$ of a given collection as shown:



$$\frac{1}{4} \text{ of } 16 = 4$$



$$\frac{1}{5} \text{ of } 25 = 5$$

Thus, to find $\frac{1}{4}$ of a collection, we divide by 4; to find $\frac{1}{5}$ of a collection, we divide by 5; and so on.



Exercise 41

1. Colour to show the fraction.

(a)



$$\frac{2}{3}$$

(b)



$$\frac{3}{4}$$

(c)



$$\frac{1}{3}$$

(d)



$$\frac{1}{4}$$

(e)



$$\frac{1}{2}$$

2. Shade one-half of the collection and write the result as shown.

(a)



$$\frac{1}{2} \text{ of } 8 = 4$$

(b)



$$\frac{1}{2} \text{ of } 10 = 5$$

(c)



$$\frac{1}{3} \text{ of } 15 = 5$$

(d)



$$\frac{1}{4} \text{ of } 20 = 5$$

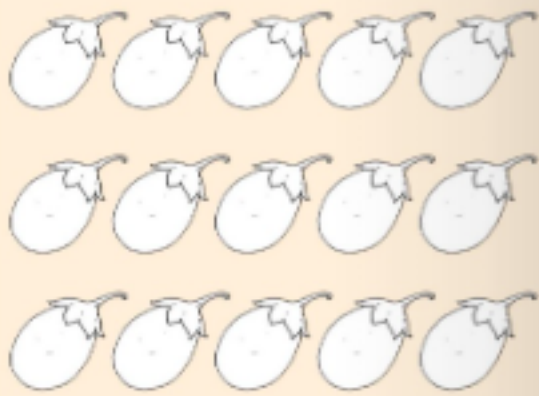
3. Shade one-third of the collection and write the result as shown. One has been done for you.

(a)



$$\frac{1}{3} \text{ of } 12 = 4$$

(b)



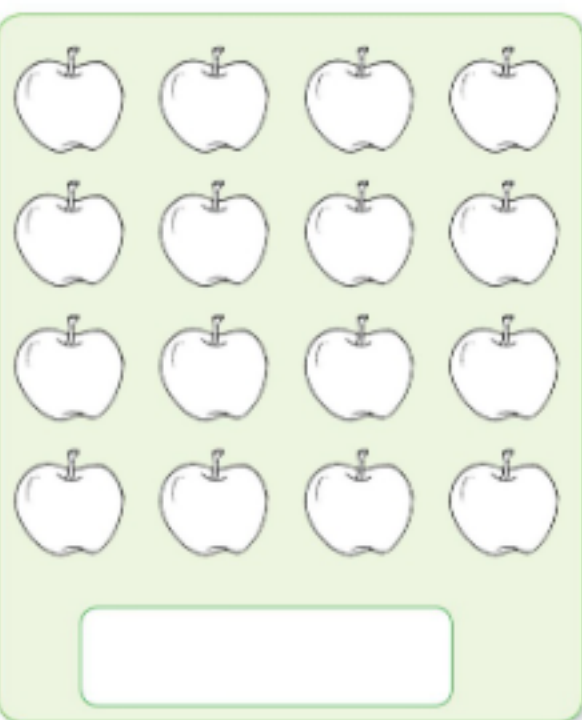
$$\frac{1}{3} \text{ of } 15 = 5$$

(c)

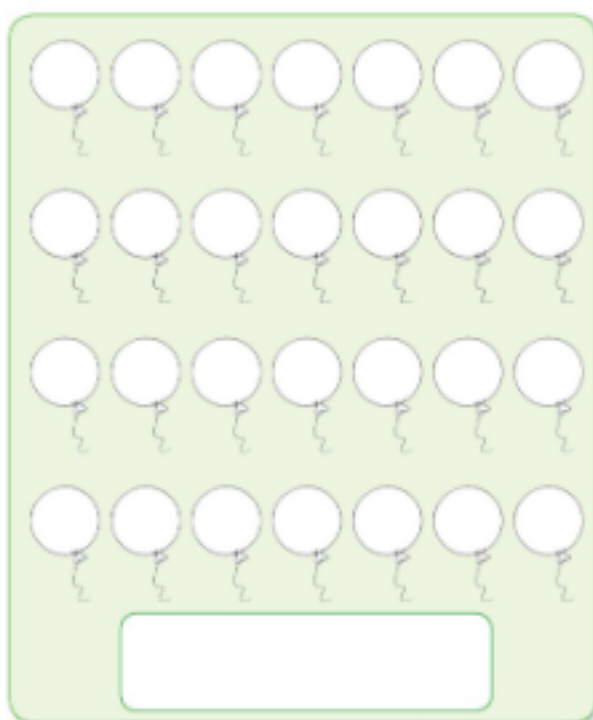


4. Shade one-fourth of the collection and write the result.

(a)



(b)



5. Solve:

(a) $\frac{1}{2}$ of 22 =

(b) $\frac{1}{3}$ of 18 =

(c) $\frac{1}{4}$ of 36 =

(d) $\frac{1}{3}$ of 27 =

(e) $\frac{1}{2}$ of 28 =

(f) $\frac{1}{4}$ of 56 =

(g) $\frac{1}{5}$ of 20 =

(h) $\frac{1}{5}$ of 45 =

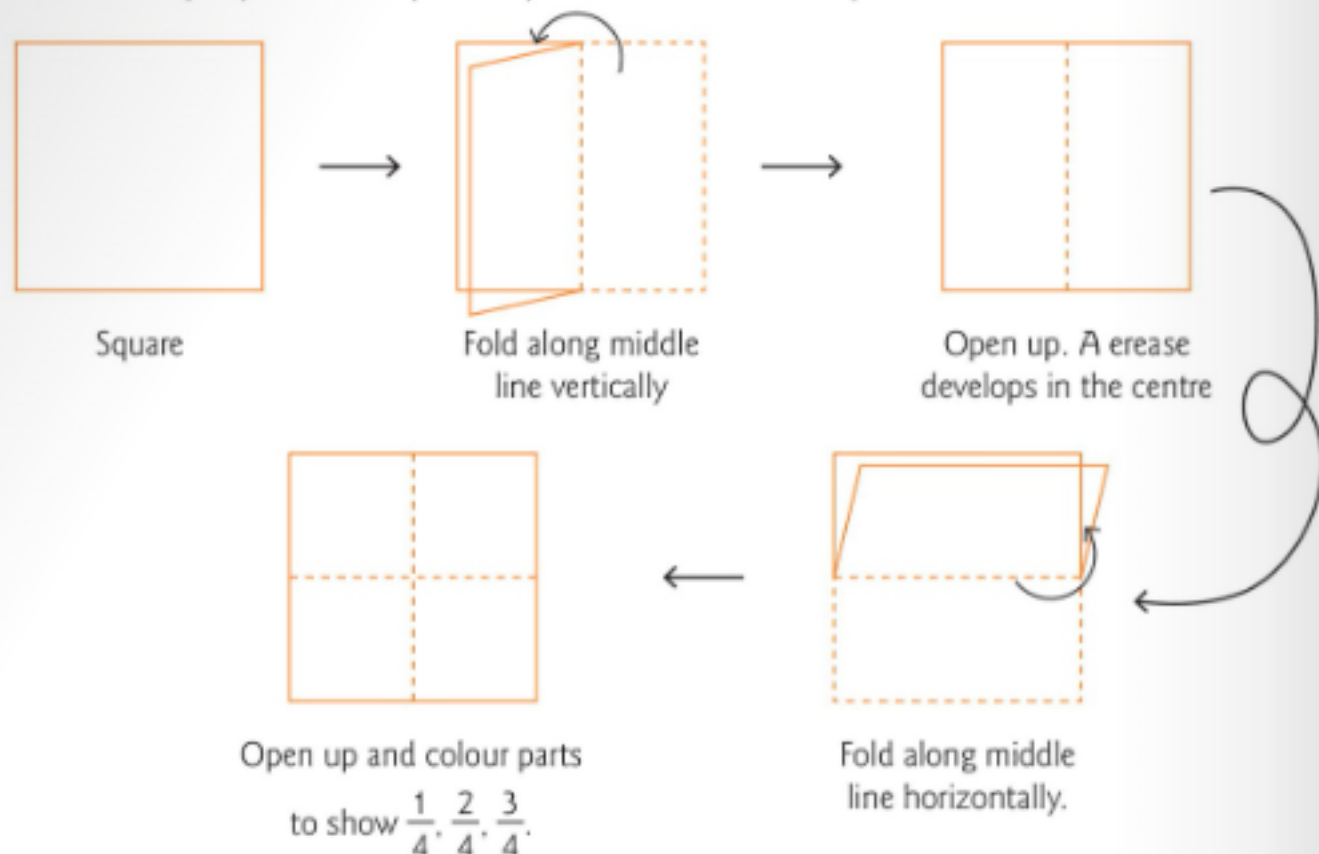


Activity Time

1. Learning Fractions By Paper Folding

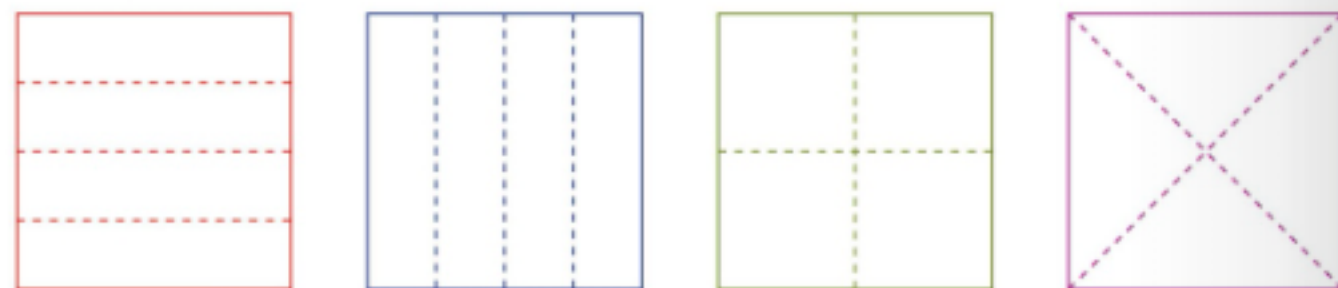
The teacher may ask students to cut out shapes of rectangles, squares, circles, triangles, stars, etc. and then fold them into equal parts.

The students may then form creases by pressing them and open them up so that the figure gets divided into equal parts. These parts may then be coloured to depict various fractions.



The above activity may be repeated with a circle and other shapes.

The teacher may also ask students to try and fold the same shape in various different ways to divide it into equal parts. An example of a square is shown below.



The students may stick these cut-outs in their notebooks or prepare charts to put them up on class display boards.

2. Learning Fraction Of A Collection The Fun Way

The teacher may ask students to draw 3×5 , 4×4 , 3×4 etc. grids on paper and then colour them differently to depict various fractions. One sample is shown below.

Red			Green	Green
Red			Green	Green
Red			Green	Green
Red			Green	Green

Clearly, in this grid, the part shaded in red shows one-fifth $\left(\frac{1}{5}\right)$ of the whole.

Also, $\frac{1}{5}$ of 20 = 4.

The part shaded in green shows two-fifths $\left(\frac{2}{5}\right)$ of the whole.

Thus, $\frac{2}{5}$ of 20 = 8.

The students may prepare numerous such grids and paste them in their notebooks.



Assessment 8

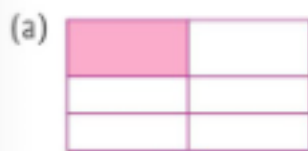
21st CS

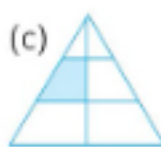
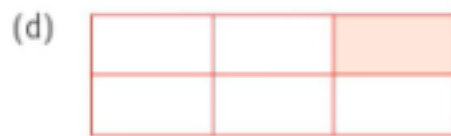
QUESTION BAG 1

(Objective Type Questions)

Tick (✓) the correct answer.

1. Which figure shows $\frac{1}{6}$ shaded?


☐

☐

☐

☐

2. What fraction of the candles on the cake is lit?

(a) $\frac{1}{4}$ ☐

(b) $\frac{3}{7}$ ☐

(c) $\frac{4}{7}$ ☐

(d) $\frac{4}{3}$ ☐



3. The figure shown alongside is shaded to show which fraction?

(a) $\frac{1}{4}$ ☐

(b) $\frac{1}{3}$ ☐

(c) $\frac{2}{3}$ ☐

(d) $\frac{3}{4}$ ☐



4. What is the fraction of each colour in a rainbow?

(a) $\frac{1}{6}$ ☐

(b) $\frac{2}{5}$ ☐

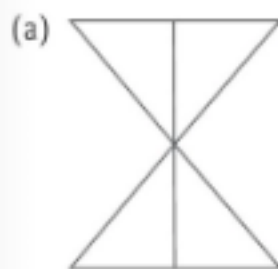
(c) $\frac{1}{7}$ ☐

(d) $\frac{2}{7}$ ☐

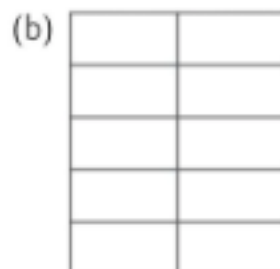
5. How many quarters will make a whole?
 (a) 2 ☐ (b) 3 ☐ (c) 4 ☐ (d) 6 ☐
6. In a bouquet, there are 4 red flowers, 6 yellow flowers and 3 white flowers. What fraction of the bouquet is red-coloured?
 (a) $\frac{4}{9}$ ☐ (b) $\frac{4}{13}$ ☐ (c) $\frac{4}{10}$ ☐ (d) $\frac{6}{13}$ ☐
7. The numerator of the fraction $\frac{13}{16}$ is
 (a) 13 ☐ (b) 16 ☐ (c) 3 ☐ (d) 6 ☐
8. What fraction of our National Flag is white?
 (a) $\frac{1}{3}$ ☐ (b) $\frac{2}{3}$ ☐ (c) $\frac{1}{2}$ ☐ (d) $\frac{3}{2}$ ☐
9. Shaloo invited 7 girls to her birthday party. Only 3 girls came. What fraction of the girls did not attend the party?
 (a) $\frac{3}{4}$ ☐ (b) $\frac{4}{3}$ ☐ (c) $\frac{3}{7}$ ☐ (d) $\frac{4}{7}$ ☐
10. I have done one-third of my homework. What fraction is left to be done?
 (a) $\frac{1}{3}$ ☐ (b) $\frac{1}{2}$ ☐ (c) $\frac{2}{3}$ ☐ (d) None of these ☐
11. In the word **ADDITION**, which fraction is represented by the letter 'D'?
 (a) $\frac{1}{8}$ ☐ (b) $\frac{2}{7}$ ☐ (c) $\frac{2}{8}$ ☐ (d) $\frac{3}{8}$ ☐
12. What is the fraction of even numbers in the given table?
 (a) $\frac{6}{12}$ ☐ (b) $\frac{5}{12}$ ☐ (c) $\frac{7}{9}$ ☐ (d) $\frac{7}{12}$ ☐
- | | | | |
|----|-----|----|----|
| 74 | 81 | 13 | 16 |
| 22 | 100 | 90 | 99 |
| 10 | 88 | 27 | 5 |
13. How many beads will make half of 10 beads?
 (a) 4 ☐ (b) 5 ☐ (c) 6 ☐ (d) 8 ☐
14. How many flowers will make one-fourth of 20 flowers?
 (a) 4 ☐ (b) 5 ☐ (c) 8 ☐ (d) 10 ☐

QUESTION BAG 2

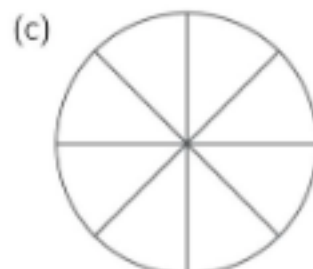
1. Shade each figure to show the given fraction:



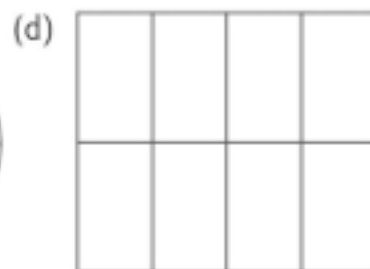
$$\frac{1}{2}$$



$$\frac{1}{5}$$



$$\frac{3}{8}$$



$$\frac{3}{4}$$

2. Fill in the blanks:

- There are one-tenths in a whole.
- There are one-sevenths in a whole.
- There are one-fifths in a whole.
- In $\frac{7}{8}$, 7 is the and 8 is the
- There are quarters in a whole.
- In a fraction tells us the total parts a figure is made of and tells us the number of parts taken.



3. If in each of the following questions, all the addends are the same, fill in the boxes.

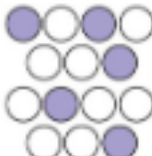
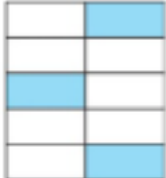
(a) + = 1

(b) + + = 1

(c) + + + = 1

(d) + + + + = 1

4. Complete the following table:

S. No.	Fraction shaded	Fraction unshaded	S. No.	Fraction shaded	Fraction unshaded
(a) 			(f) 		
(b) 			(g) 		
(c) 			(h) 		
(d) 			(i) 		
(e) 			(j) 